Michael Capera

Software Engineer

Colombia
linkedin.com/in/michael-capera-47ab0a18a | michaelcapera.github.io/Portfolio-Michael_Capera

PROFESSIONAL SUMMARY

Software Engineer with 8+ years of experience, including fintech and banking environments, specializing in cloud-native full-stack development on AWS. I build secure, scalable serverless systems, automate infrastructure with Terraform and CI/CD, and apply Spec-Driven and Test-Driven Development to deliver reliable, well-documented software for financial operations.

SKILLS

Cloud & DevOps: AWS (Lambda, API Gateway, EC2, S3, ECS, RDS, DynamoDB), Azure, Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, GitLab CI, Jenkins, Bitbucket Pipelines).

Backend & APIs: Python (Django, Flask, FastAPI, Celery), Java (Spring Boot), Node.js (Express), microservices architecture, RESTful APIs, GraphQL, security (OAuth, JWT).

Frontend: React, Angular, Vue.js, TypeScript, JavaScript, HTML5, CSS3, Redux, Context API, Stencil.js (web components), Jest / Enzyme.

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Memcached.

AI & Data: Amazon Bedrock, Azure Language Studio, NLP, computer vision, machine learning, Pandas, Selenium, Webots.

Engineering Practices: Spec-Driven Development (SDD), Test-Driven Development (TDD), unit testing, Infrastructure as Code, API documentation, system design.

Fintech: payment processing, security & compliance (GDPR, PCI-DSS), high-availability systems.

Languages: Spanish (native), English (professional proficiency).

WORK HISTORY

Software Engineer
2026 – Present

COHNECTI — ROCA ERP

- Develop Roca ERP, a cloud-native ERP platform, with a Python / Django backend and a React JS frontend.
- Applied Spec-Driven Development (SDD), defining features as specifications before implementation to keep product, code and tests aligned.
- Scaled core monolithic workflows by moving heavy processes to Celery workers running in Docker containers on AWS ECS.
- Automated infrastructure with Terraform and CI/CD with Bitbucket Pipelines, reducing deployment time by 40%.
- Built serverless microservices with AWS Lambda (Python) and API Gateway, backed by RDS and DynamoDB.

Software Engineer
2021 – 2025

BANCO DE BOGOTÁ

- Designed and developed serverless, high-availability applications with AWS Lambda (Node.js / Python), API Gateway and AppSync (GraphQL) for critical banking operations.
- Built reusable web components with Stencil.js to standardize and accelerate UI development across the organization.
- Managed high-volume DynamoDB and RDS databases, ensuring data integrity and performance for financial transactions.
- Improved code quality through unit testing and TDD, and created comprehensive API documentation.
- Provisioned cloud resources as code with Terraform.

Software Engineer 2021

CASA EDITORIAL EL TIEMPO

- Developed and optimized high-traffic websites and landing pages with HTML, CSS, JavaScript and PHP.
- Delivered fully responsive, cross-browser interfaces, improving user experience and accessibility.
- Worked closely with UX/UI designers to turn complex wireframes into usable, visually appealing interfaces.

Software Engineer 2021

AGRAPP

- Built key application features with Angular on a modern client-side architecture.
- Integrated GraphQL through Apollo Client for efficient, optimized data fetching.
- Used DynamoDB and AWS APIs for data storage and application features.

Freelance Software Engineer 2020 – 2025

REAL ESTATE & AUTOMATION SOLUTIONS

- **Real estate automation:** developed a credit and property analysis platform in Python.
- **Document processing:** automated generation and parsing of legal real estate documents and credit reports with PyMuPDF.
- **Web scraping & data analysis:** built Selenium scrapers to aggregate property market data and analyzed trends with Pandas.
- **Automated marketing:** generated publicity images automatically by integrating Python with Canvas templates, streamlining lead generation.
- **Agentic AI & NLP:** automated contact form processing and lead qualification with NLP on AWS Lambda and API Gateway.
- **Security:** added automated file scanning and threat detection with the VirusTotal API.

Backend Developer Mar 2019 – Mar 2020

MEDICAL APP

- Led full-cycle backend development of a web application, managing data flow and application logic.
- Implemented the MVC pattern for a clean separation of concerns and maintainable code.
- Developed features with PHP and JavaScript, using jQuery for dynamic frontend interactions.
- Designed and optimized the database for efficient data access.

Web Developer Feb 2018 – Mar 2019

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- Built and maintained web applications with PHP, JavaScript, HTML and CSS.
- Applied the MVC pattern to structure code, improving scalability and easing future development.
- Enhanced user interfaces and interactivity with jQuery.
- Helped define technical requirements and implemented solutions for the client's business needs.

PERSONAL PROJECTS

- Self-driving car using deep learning and computer vision.
- Agentic AI and natural language processing (NLP).
- Machine learning and Markov models with Python.

EDUCATION & TRAINING

Software Engineering — Politécnico Grancolombiano 2017 – Present

Developing on AWS — Amazon Web Services